

A – Multiple Choice

Q	Ans	Q	Ans	Q	Ans
1	b	2	c	3	b
4	a	5	c	6	b
7	c	8	b	9	a
10	a	11	c	12	c
13	b	14	b	15	b
16	c	17	a	18	a
19	c	20	a		

B – Short Answer

(21) All subsets of $\{a, b, c\}$:

$\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} - 8$ in total.

(24) (i) $A \cap B = \{2, 6\}$

(ii) $A \nabla B = (A \cup B) \setminus (A \cap B) = \{0, 1, 2, 3, 4, 5, 6\} \setminus \{2, 6\} = \{0, 1, 3, 4, 5\}$

(22) $\{5, 6, 7, 8\}$

(23) $D = \{x : x = 4n, n \in \mathbb{N}\}$

(19) Power set has $2^n = 2^5 = 32$ elements.

(20) Complement of complement returns the set: $(A')' = A$

C – Problems

(25) $n(F \cup B) = 35 + 25 - 12 = 48$. (a) Exactly one = $35 + 25 - 24 = 36$. (b) Neither = $60 - 48 = 12$. (c) Venn: football-only 23, both 12, basketball-only 13, outside 12.

(26) (a) $A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}$ (b) $A \cap B = \{2, 4\}$ (c) $A \setminus B = \{6, 8, 10\}$ (d) $B \setminus A = \{1, 3, 5\}$ (e) $n(A) = 5, n(B) = 5$, so $n(A \cup B) = 5 + 5 - 2 = 8 \checkmark$.

(27) $U = \{1, \dots, 12\}, A = \{1, 3, 5, 7, 9\}, B = \{2, 3, 5, 7, 11\}$.

(a) $A' = \{2, 4, 6, 8, 10, 11, 12\}$

(b) $A \cap B = \{3, 5, 7\}$

(c) $A \cup B = \{1, 2, 3, 5, 7, 9, 11\}$

(d) $(A \cup B)' = \{4, 6, 8, 10, 12\}$